



Solarowl Energy Solutions Private Limited

EV CHARGERS

Complete Charging Solutions



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Solarowl Energy EV Solutions is the electric vehicle supply equipment solution arm Solarowl Energy Solutions PVT. LTD. Solarowl Energy was formed in 2013 and is one of the leading manufacturing companies in India with operations across embedded and power electronics, solar energy, industrial automation, pumps, and motors.

Solarowl Energy EV Solutions has a strong research and development team dedicated to design state of the art indigenous EV chargers as per Indian and Global market requirements.

Solarowl Energy EV Solutions currently offers an energy efficient range of EV Chargers including AC Chargers, DC Chargers and Combination of AC & DC Chargers as per Indian and International standards. We also offer software-as-a- service Central Management System solutions to operate charger networks. These comprehensive EV charging solutions are able to fulfill the needs for various applications such as parking, corporate buildings, fleets, and residential buildings.

Solarowl Energy EV Solutions offers a wide range of AC chargers up to 42 kW in AC Type-2, 10 kW in BHARAT AC 001, and up to 240 kW in DC Chargers. We also offers combined 2-in-1 and 3-in-1 chargers which includes a combination of CCS-2, CHAdeMO and AC Type-2.

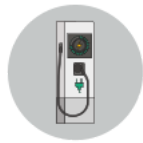
APPLICATIONS



HIGHWAY



PARKING



SERVICE
STATION



COMMERCIAL



FLEET



RESIDENTIAL



HOTEL

FEATURES

- AC Type-2 complies with IEC 62196 and BHARAT AC001 complies IEC 60309.
- CCS-2 complies with IEC 62196-3, CHAdeMO complies with JEVS G104 and BHARAT DC001 with GB/T 20234.2.
- 3 Gun in 1 Charger : Output power capacity up-to 120 kW for CCS-2, CHAdeMO & 43kW for AC Type-2
- 2 Gun in 1 Charger : Output power capacity up-to 30 kW for CCS-2, CHAdeMO & 15 kW for DC001 (Each gun)
- 1 Gun in 1 Charger : Output power capacity up-to 240 kW for CCS-2 single gun ultra fast charger.
- Simultaneous charging on each gun with load sharing for 2 guns and 3 guns variant DC fast EV chargers.
- High efficiency of >99% and >95% respectively for AC and DC chargers.
- Available models in 1-Ph and 3-Ph power inputs.
- High brightness HMI color touch screen of 4.3" or 7" as per the charger capacity and size.
- Front panel visual indication for Mains available, System error, Charging status
- Suitable for extreme Indian environmental conditions.
- Payment options integration with RFID, payment wallets, and credit card
- Modular design resulting in streamlined support and service
- All-weather enclosure and HMI for use inside or outside (IP54)
- Built-in communication modules for (3G/Wifi/LAN)
- Compliance with OCPP 1.6J protocol

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Product Capacity Range :-

			
Parameter	AC Type-2	BHARAT AC001	CCS-2
Input Voltage (V AC)	1-Ph 230V or 3-Ph 400V (+/- 10%)	3 Phase 415V (375-440 Vac) as per IS 12360	3-Ph 400V (+/- 10%)
Efficiency (%)	>99.9%	>99.9%	>= 94%
Input Frequency (Hz)	50 Hz	50 Hz	45-65 Hz
Wires	3 wire (L,N,PE) or 5 wire (L1,L2,L3,N,PE)	5 wire (L1,L2,L3,N,PE)	5 wire (L1,L2,L3,N,PE)
Number of Outputs	1	3	1
Output Connector type	IEC 62196 TYPE-2 AC	IEC 60309	IEC 62196-3
Output Voltage (V)	1-Ph 230V or 3-Ph 400V AC	1-Ph 230V AC	100-750V DC
Max. Output Current (A AC/DC)	Max. 62A AC	Max. 15A AC	Max. 80A DC
Max. Output Power Rating (kW)	7kW, 11kW, 22kW, 43kW AC	9.9 kW AC (3.3 kw for each Gun)	Max. 30 kW DC
Display	4.3" TFT Color Touch Screen	3.5" Inch Color Touch Screen	4.3" TFT Color Touch Screen
Emergency Push Button	Provided	Provided	Provided
User Authentication	RFID based	RFID based	RFID based
Visual Indication	LCD and LED for Power, Charge and Fault	LCD and LED for Power, Charge and Fault	LCD and LED for Power, Charge and Fault
Charging Mode	IEC 61851-1	-	IEC 61851-1 (Mode-4)
Charger and Vehicle Communication	As per IEC 61851-1	-	Power Line Communication (PLC)
Charger and CMS communication	Open Charge Point Protocol 1.6J	Open Charge Point Protocol 1.6J	Open Charge Point Protocol 1.6J
Charger Interface	Ethernet, 3G/4G, Wifi	Ethernet, 3G/4G, Wifi	Ethernet, 3G/4G, Wifi
Protections	Input Over Voltage, Input Under Voltage, Connection Abnormality, Emergency Stop, Lightning Protection, Leakage, Short Circuit, Over Current, Over Temperature, Reverse Battery Connection		
Ambient Temperature	-20 to +50 C	-20 to +50 C	-20 to +50 C
Place of Installation	Indoor and Outdoor	Indoor and Outdoor	Indoor and Outdoor
Ingress Protection	IP54	IP54	IP54
Enclosure Protection	-	IK10	IK10
Cooling	Natural Convection	Natural Convection	Natural Convection
Charging wire length	5 mtr	5 mtr	5 mtr
Max. dimension (W X D X H) mm	760 X 600 X 1620	283 X 185 X 400	530 X 270 X 752
Max. Weight (kGs)	75 kGs	7 kGs	68 kGs

			
Parameter	CHAdMO	BHARAT DC001	CCS-2 (Ultra Fast)
Input Voltage (V AC)	3-Ph 400V (+/- 10%)	3-Ph 400V (+/- 10%)	3-Ph 400V (+/- 10%)
Efficiency (%)	>= 94%	>= 95%	>= 95%
Input Frequency (Hz)	45-65 Hz	50 Hz	45-65 Hz
Wires	5 wire (L1,L2,L3,N,PE)	5 wire (L1,L2,L3,N,PE)	5 wire (L1,L2,L3,N,PE)
Number of Outputs	1	1 or 2	1
Output Connector type	JEVS G104 (CHAdMO)	GB/T 20234.3	IEC 62196-3 (CCS-2)
Output Voltage (V)	100-500V DC	20-100V DC	200-1000V DC (CCS-2)
Max. Output Current (A AC/DC)	Max. 80A DC	Max. 200A DC	Max. 400A
Max. Output Power Rating (kW)	Max. 30 kW DC	Max 30 kW DC	Max. 240 kW DC
Display	4.3" TFT Color Touch Screen	7" TFT Color Touch Screen	7" TFT Color Touch Screen
Emergency Push Button	Provided	Provided	Provided
User Authentication	RFID based	OTP/RFID based	OTP/RFID based
Visual Indication	LCD and LED for Power, Charge and Fault	LCD and LED for Power, Charge and Fault	LCD and LED for Power, Charge and Fault
Charging Mode	IEC 61851-1 (Mode-4)	-	IEC 61851-1 (Mode-4) for DC
Charger and Vehicle Communication	CAN	CAN	Power Line Communication (PLC)
Charger and CMS communication	Open Charge Point Protocol 1.6J	Open Charge Point Protocol 1.6J	Open Charge Point Protocol 1.6J
Charger Interface	Ethernet, 3G/4G, Wifi	Ethernet, 3G/4G, Wifi	Ethernet, 3G/4G, Wifi
Protections	Input Over Voltage, Input Under Voltage, Connection Abnormality, Emergency Stop, Lightning Protection, Leakage, Short Circuit, Over Current, Over Temperature, Reverse Battery Connection		
Ambient Temperature	-20 to +50 C	-20 to +50 C	-20 to +50 C
Place of Installation	Indoor and Outdoor	Indoor and Outdoor	Indoor and Outdoor
Ingress Protection	IP54	IP54	IP54
Enclosure Protection	IK10	IK10	IK10
Cooling	Natural Convection	Natural Convection	Natural Convection
Charging wire length	5 mtr	5 mtr	5 mtr
Max. dimension (W X D X H) mm	530 X 200 X 752	800 X 1050 X 1750	900 X 1010 X 1970
Max. Weight (kGs)	66 kGs	200 kGs	350 kGs

Product Capacity Range :-



Parameter	CCS-2 + CHAdeMO + AC Type-2
Input Voltage (V AC)	3-Ph 400V (+/- 10%)
Efficiency (%)	>= 95%
Input Frequency (Hz)	45-65 Hz
Wires	5 wire (L1,L2,L3,N,PE)
Number of Outputs	3
Output Connector type	IEC 62196-3 (CCS-2), JEVS G104(CHAdeMO), IEC 62196 (TYPE-2 AC)
Output Voltage (V)	100-750V DC (CCS-2), 100-500V DC (CHAdeMO), 3-Ph 400V (TYPE-2 AC)
Max. Output Current (A AC/DC)	Max. 160A DC, Max. 63A AC
Max. Output Power Rating (kW)	Max. 120 kW DC, Max. 43 kW AC
Display	7" TFT Color Touch Screen
Emergency Push Button	Provided
User Authentication	OTP/RFID based
Visual Indication	LCD and LED for Power, Charge (1/2/3 Guns) and Fault
Charging Mode	IEC 61851-1 (Mode-4) for DC, IEC 61851-1 (Mode-3) for AC
Charger and Vehicle	CAN for CHAdeMO, PLC for CCS, PWM for Type-2 AC
Communication Charger and CMS	Open Charge Point Protocol 1.6j
communication Charger Interface	Ethernet, 3G/4G, Wifi
Protections	Input Over Voltage, Input Under Voltage, Connection Abnormality, Emergency Stop, Lightning Protection, Leakage, Short Circuit, Over Current, Over Temperature, Reverse Battery Connection
Ambient Temperature	-20 to +50 C
Place of Installation	Indoor and Outdoor
Ingress Protection	IP54
Enclosure Protection	IK10
Cooling	Natural Convection
Charging wire length	5 mtr
Max. dimension (W X D X H) mm	800 X 1050 X 1750
Max. Weight (kGs)	260 kGs

All pictures shown are for illustration purpose only. Actual product may vary due to product enhancement



SOLAROWL ENERGY EV SOLUTIONS

+91 9422980148 / +91 7020660967 / +91 7040506295

www.solarowl.co.in

Product Improvement is a continuous process at 'EVenergy'. The data given in this publication is therefore subject to revision.

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